Rev 4
6/11/26

Design Rulebook

Project Goals

Generate 3D printable electromagnetic circuits in OpenSCAD. Circuits will consist of a protopasta conductive PLA trace layer and an insulating layer that separates the traces.

Physical Laws

- Conductive trace layer resistivity (ρ): 15 Ω*cm.
- Resistance: $R = \frac{\rho A_C}{L}$.
- A_C = cross-sectional area of material.
- L = length of material.

Design Principles

Rules listed below are to be consulted differently for each circuit element.
Circuit elements consist of:

- DIP components
- Passive components
- Active components
- Routing
- Insulating layer

Design Hierarchy

1. Decide on the component requirements for the circuit
    a. Components necessary, component values
2. Understand which component pins have common nodes
3. Decide where to place component groups
4. Place routing with open space for components to be installed later

Passive Components Using Conductive Trace Geometry

- Minimum conductive trace width: 1mm.
- Minimum conductive trace spacing: 1mm.
- Conductive trace height: 1mm.
- Only refer to allowed geometry styles when constructing components
- All serpentine bending should be constructed using 45-degree chamfered segments

Placement

- Components with common nodes should be placed in a close group, prioritized by the number of common nodes
- Ensure common components are separated from other component groups
- Minimum routing distance for shared nodes: 4mm
- Minimum space between distinct components: 2mm

Routing

- Maximum route bending angle: 45 degrees.
- Vias are to be utilized to pull conductive traces out of plane when a passover is required to avoid shorting
- Via minimum height: 2mm
- Via minimum radius: 1mm
- Minimum spacing between traces: 1mm
- 4 mm by 4 mm contact pads to be placed at the power and ground ends of the circuit, with full contact between the end of the trace and the contact pad.
- Contact pads should be perfectly insulated on each side from other conductive trace elements, except the trace end to which they are connected.

Insulating Layer

- The insulating layer and Conductive Trace layer are generated separately, not as a single OpenSCAD project.
- The insulating layer should provide 2mm of clearance on each trace boundary
    - i.e., the trace footprint outline is 25.4 mm by 25.4 mm; the insulating layer footprint should be 29.4 mm by 29.4 mm, matching the general shape of the trace footprint outline.

Allowed Geometry Styles for Passive Components

Resistors

- Serpentine

Capacitors

- Interdigitated

Inductors

- Planar

Optimization Goals

N/A

Style Preferences

- The insulating layer serves as a flat platform for conductive traces to be printed on.
- Only create walls in the insulating layer if necessary to isolate distinct trace planes.